Instructions for Disassembling a GT1/GT2 Steering Column and Installing Bearings

Scooter Steering Column Disassembly and Bearing Installation Guide

Required Materials and Tools:

- **Tools:**
  - Hex keys and screwdrivers set
  - Wrenches
  - Large sledgehammer (at least 3 kg)
  - Angle grinder with metal cutting disk
  - Heat gun (capable of reaching temperatures around 500°C)
  - M10 threaded rod and 3-4 nuts
  - Wide metal washers for the threaded rod
  - Wooden or plywood spacers (inner diameter 31-32 mm, thickness 10-15 mm)
  - Metal or wooden spacer for hammering
- **Consumables:**
  - 2 SKF 32006 bearings
  - Thick bearing grease
  - Silicone grease (for treating seating surfaces)
  - Fine-grit sandpaper (P1000 grit)
  - Heavy-duty adhesive tape
  - Clean wipes (for removing dust and chips)
- **Auxiliary Equipment:**
  - Spiral-cut base of an old lower bearing (used for pressing new bearings)
  - Stable supports or wooden boards (to hold the scooter firmly in position)
  - Assistant (for tasks like hammering out the steering shaft)

1. **Workspace Preparation** Begin by properly organizing your workspace. The task is extensive, complicated, and labor-intensive, so stability is crucial. Securely position the scooter—boards were used, for instance, to ensure it doesn't tip over and remains firmly in place.
2. **Disassembly – Top to Bottom**
   - Begin by dismantling the scooter from the top down, starting with brake lines and wires.
   - Disconnect the brake lines and seal them thoroughly with adhesive tape.
   - Also, seal the holes on the brake levers using the same tape.
3. **Steering Column Disassembly**
   - Unscrew and remove the upper handlebar part after disconnecting the internal cable within the steering column.
   - Unscrew one bolt on the folding mechanism of the handlebars to unfold it and access the large internal screw inside the steering column.
4. **Removing Charging Ports**
   - Unscrew the plastic mounting for the charging ports and remove them together with the mounting.
   - This allows side access to the steering column, where cables and the rear brake line pass.
5. **Accessing the Controller**
   - Unscrew and remove the scooter deck cover that encloses the main controller.
   - This step ensures wires and the brake line won't kink and can freely exit through the hole near the controller.
6. **Removing Front Brake System**
   - Unscrew the front brake caliper—this facilitates pulling out the front brake line.
   - Completely remove the front brake line along with the caliper.
7. **Extracting Wires and Rear Brake Line**
   - Carefully pull out wires and the rear brake line from the steering column through the hole near the controller.
   - Note: soft plastic decorative covers on brake lines (covering bolts near handles) should not be cut—they fit through all holes.
8. **Steering Column Removal**
   - Unscrew all bolts securing the steering column:
     - One large internal screw inside the column.
     - Several smaller external screws.
     - The centering screw.
   - Remove the front suspension along with shock absorber and wheel.
   - Remove the grey aluminum decorative overlay, exposing the aluminum frame.
9. **Preparation for Steering Shaft Removal**
   - After these steps, only the steering shaft with two bearings should remain in the column.
   - Screw the large galvanized bolt fully into the steering shaft.
10. **Hammering Out the Steering Shaft**

- Required:

    - Heat gun (500°C).
    - Spacer for striking (metal or dense wood).
    - Assistant.
  - Heat the upper bearing evenly from all sides, holding the heat gun nozzle 1.5–2 cm away. Smoke from leftover bearing grease indicates readiness.
  - Set aside the heat gun, place the spacer on the galvanized bolt.
  - The assistant hits the spacer hard with a sledgehammer—two strikes were sufficient to remove the shaft.
  - **Important:** Don't use a regular hammer—risk of damage. Only a large sledgehammer is recommended, needing just a few hits.

11. **Removing the Lower Bearing**

  - After extracting the shaft, the lower bearing remains attached.
  - Cut it carefully using the angle grinder, avoiding damage to the seating surface.
  - Steps:
    - First, cut the bearing cage.
    - Then spiral-cut to loosen the bearing.
    - Under tension, it will crack and can then be pried off with a screwdriver.

Installation (Pressing) of New Bearings into the Scooter's Steering Column

1. **Required Materials and Tools:**
    - 2 new `SKF 32006` bearings;
    - Thick bearing grease;
    - Spiral-cut base from the old lower bearing (mandatory! Risk of bearing damage without it);
    - Wooden or plywood washer spacers (to press bearings into place). Alternatives like plumbing fixtures might work but were not tested.
2. **Preparing Seating Surfaces:**
    - Carefully smooth bearing seating surfaces using very fine sandpaper (P1000).
    - Remove any remaining dust or chips using clean wipes.
    - Apply a small amount of silicone grease to both upper and lower seating surfaces.
3. **Pressing in the Lower Bearing:**
    - Place the new lower bearing onto the steering shaft.
    - Immediately follow with the spiral-cut old bearing base (important! the old base must press against the new bearing's narrow part).
    - Next, position thick wooden or plywood washer spacers.
    - Gradually press the bearing downward by tightening the large galvanized bolt.
    - Initially, the bearing will settle into the first seat.
    - Gradually add more spacers to push the bearing deeper.

- Important: Ensure the bearing isn't pressed too far—seating must be precise.

4. **Installing Bearing Races in the Steering Column:**
   - Use:
     - Metal washers;
     - M10 threaded rod with two lock nuts.
   - First, install the lower bearing race.
   - Thread the rod with washers through the lower bearing race and column from below.
   - Place atop the rod:
     1. Wooden washer (to protect the scooter's aluminum);
     2. Metal washer;
     3. Tighten the nut on the rod.
   - Tightening the top nut gently pulls the lower bearing race into position.
   - Repeat this procedure for the **upper** bearing race.
5. **Installing the Steering Shaft and Upper Bearing:**
   - Insert the steering shaft with the installed lower bearing into the steering column.
   - Slide on the upper bearing.
   - Immediately add the same spiral-cut old bearing base.
   - Using the same thick wooden spacers from the lower bearing installation, tighten the top bolt.
   - **Important:** Don't fully tighten the bearing—leave slight technological clearance. Final tightening during assembly will eliminate this clearance without risking over-tightening and jamming.
6. **Completing Assembly:**
   - After pressing all components, assemble the steering column in reverse order, exactly repeating disassembly steps.

Note: Minor operations are omitted here, assuming familiarity with general procedures—such as installing rubber plugs or gaskets—which should be obvious to anyone performing this task.

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